

# Lab 2 – My Solutions

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## Task 1 – Create User Accounts in Active Directory

**Step 4 – Create John Doe via GUI** (*Done manually in Active Directory Users and Computers - no command needed*)

## Step 9 – PowerShell: Find and use the AD user creation command

```
powershell
# Find the right command
Get-Command *User

# Get help for New-ADUser
Get-Help New-ADUser
Get-Help New-ADUser -Examples
Get-Help New-ADUser -Online
```

## Create several classmate accounts via PowerShell:

```
powershell
# Import AD module (if not already loaded)
Import-Module ActiveDirectory

# Create user: Jane Doe (same as GUI step 8, but via PowerShell)
New-ADUser `
  -Name "Jane Doe" `
  -GivenName "Jane" `
  -Surname "Doe" `
  -SamAccountName "janed" `
  -UserPrincipalName "janed@contoso.com" `
  -DisplayName "Jane Doe" `
  -Description "Lab 2 user - Jane Doe" `
  -AccountPassword (ConvertTo-SecureString "P@ssw0rd" -AsPlainText -Force) `
  -Enabled $true `
  -MemberOf "Domain Users"

# Create user: Alice Smith
New-ADUser `
  -Name "Alice Smith" `
  -GivenName "Alice" `
  -Surname "Smith" `
  -SamAccountName "asmith" `
  -UserPrincipalName "asmith@contoso.com" `
  -DisplayName "Alice Smith" `
  -Description "Lab 2 user - Alice Smith" `
  -AccountPassword (ConvertTo-SecureString "P@ssw0rd" -AsPlainText -Force) `
  -Enabled $true
```

```
# Create user: Bob Brown
New-ADUser `
  -Name "Bob Brown" `
  -GivenName "Bob" `
  -Surname "Brown" `
  -SamAccountName "bbrown" `
  -UserPrincipalName "bbrown@contoso.com" `
  -DisplayName "Bob Brown" `
  -Description "Lab 2 user - Bob Brown" `
  -AccountPassword (ConvertTo-SecureString "P@ssw0rd" -AsPlainText -Force) `
  -Enabled $true
```

## Step 10 – Change a property via GUI, then verify in PowerShell:

```
powershell
# View all properties of a specific user
Get-ADUser -Identity "jdoe" -Properties *

# View specific properties
Get-ADUser -Identity "jdoe" -Properties DisplayName, Description, WhenCreated
```

## Step 11 – Filter and list users:

```
powershell
# List all users with basic properties
Get-ADUser -Filter * -Properties DisplayName, Description | Select Name, DisplayName, Description

# Filter users whose DisplayName contains "Doe"
Get-ADUser -Filter {DisplayName -like "*Doe*"} -Properties DisplayName | Select Name, DisplayName

# Filter users created in the last 7 days
$since = (Get-Date).AddDays(-7)
Get-ADUser -Filter {WhenCreated -ge $since} -Properties WhenCreated | Select Name, WhenCreated
```

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## Task 2 – Create Local User Accounts (on CL1)

### Find the local user creation command and create accounts:

```
powershell
# Find the command
Get-Command *LocalUser

# Create local user: John Local (as per GUI step 3)
New-LocalUser `
  -Name "johnlocal" `
  -FullName "John Local" `
  -Description "Local account - John Local" `
  -Password (ConvertTo-SecureString "P@ssw0rd" -AsPlainText -Force) `
  -PasswordNeverExpires

# Add to local Users group
```

```

Add-LocalGroupMember -Group "Users" -Member "johnlocal"

# Create more local classmate accounts
New-LocalUser `
  -Name "alicelocal" `
  -FullName "Alice Local" `
  -Description "Local account - Alice" `
  -Password (ConvertTo-SecureString "P@ssw0rd" -AsPlainText -Force) `
  -PasswordNeverExpires

Add-LocalGroupMember -Group "Users" -Member "alicelocal"

New-LocalUser `
  -Name "boblocal" `
  -FullName "Bob Local" `
  -Description "Local account - Bob" `
  -Password (ConvertTo-SecureString "P@ssw0rd" -AsPlainText -Force) `
  -PasswordNeverExpires

Add-LocalGroupMember -Group "Users" -Member "boblocal"

# Verify local users were created
Get-LocalUser

```

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## Task 3 – Create Groups in Active Directory

### Step 1 & 2 done via GUI. Step 3 via PowerShell:

```

powershell
# Find the AD group creation command
Get-Command *ADGroup

# Create Lab2 Group (same as GUI)
New-ADGroup `
  -Name "Lab2 Group" `
  -SamAccountName "lab2group" `
  -GroupScope Global `
  -GroupCategory Security `
  -Description "Group for Lab 2"

# Add John Doe to Lab2 Group
Add-ADGroupMember -Identity "lab2group" -Members "jdoe"

# Create additional groups for classmates
New-ADGroup `
  -Name "ClassmatesGroup" `
  -SamAccountName "classmatesgroup" `
  -GroupScope Global `
  -GroupCategory Security `
  -Description "Group for classmates"

# Add multiple users to the group

```

```
Add-ADGroupMember -Identity "classmatesgroup" -Members "asmith", "bbrown", "janed"
```

```
# Verify group members
```

```
Get-ADGroupMember -Identity "lab2group" | Select Name, SamAccountName
```

```
Get-ADGroupMember -Identity "classmatesgroup" | Select Name, SamAccountName
```

---

## Task 4 – Create Local Groups (on CL1)

```
powershell
```

```
# Find local group commands
```

```
Get-Command *LocalGroup
```

```
# Create first local group via PowerShell
```

```
New-LocalGroup -Name "Lab2LocalGroup" -Description "Local group for Lab 2"
```

```
# Add local users to it
```

```
Add-LocalGroupMember -Group "Lab2LocalGroup" -Member "johnlocal"
```

```
Add-LocalGroupMember -Group "Lab2LocalGroup" -Member "alicelocal"
```

```
# Create second group (the first was via Computer Management GUI)
```

```
New-LocalGroup -Name "Lab2LocalGroup2" -Description "Second local group for Lab 2"
```

```
Add-LocalGroupMember -Group "Lab2LocalGroup2" -Member "boblocal"
```

```
# Verify
```

```
Get-LocalGroup
```

```
Get-LocalGroupMember -Group "Lab2LocalGroup"
```

---

## Task 5 – Administrators

### Add John Doe to Domain Admins (on DC1 via PowerShell):

```
powershell
```

```
# Add jdoe to Domain Admins
```

```
Add-ADGroupMember -Identity "Domain Admins" -Members "jdoe"
```

```
# Verify
```

```
Get-ADGroupMember -Identity "Domain Admins" | Select Name
```

### Verify jdoe can create a local user on CL1 (log in as contoso\jdoe, then run):

```
powershell
```

```
New-LocalUser `
```

```
-Name "testadmin" `
```

```
-FullName "Test Admin Account" `
```

```
-Password (ConvertTo-SecureString "P@ssw0rd" -AsPlainText -Force) `
```

```
-PasswordNeverExpires
```

```
# jdoe has admin rights 
```

**"Sign in to DC1 as John Doe and Jane Doe. Is the result the same? Why?"**

**John Doe** (jdoe) is a member of **Domain Admins**, which gives him rights to log on locally to domain controllers. He can sign in to DC1 and can create new user accounts.

**Jane Doe** (janed) is only a regular **Domain User**. By default, regular domain users are **not allowed to log on locally to a Domain Controller** (DC). DCs restrict interactive logon to members of specific groups (Domain Admins, Administrators, etc.) via the *"Allow log on locally"* Group Policy setting. Jane's login to DC1 will **fail**.

**Can John create new user accounts on DC1?** ☒ Yes - he's a Domain Admin.

**Can Jane create new user accounts on DC1?** ☐ No - she can't even log in, and lacks the permissions.